BIOL 3400 — Probability 2 Worksheet

Bayes' Theorem: HIV Testing, Sequential Updates & Mammography

Exercise 1: HIV Testing and Bayes' Theorem

0.3% of Americans have HIV. The HIV test has a false negative rate of 1% and a false positive rate of 5%.

Step 1: List what you know

P(A) = HIV present	_____	*prevalence*
P(~A) = HIV absent	_____	*1 − prevalence*
P(B|A) = Test+ given HIV	_____	*sensitivity = 1 − false negative rate*
P(~B|A) = Test− given HIV	_____	*false negative rate*
P(B|~A) = Test+ given no HIV	_____	*false positive rate*
P(~B|~A) = Test− given no HIV	_____	*specificity = 1 − false positive rate*

Step 2: Question 1 — What is P(Test+)?

P(B) = P(B|A) × P(A) + P(B|~A) × P(~A)

= _____ × _____ + _____ × _____

= _____

Tests are positive about _____% of the time.

Step 3: Question 2 — P(HIV | Test+)?

Apply Bayes' theorem:

P(A|B) = P(B|A) × P(A) / P(B)

= _____ / _____

= _____

_____% of individuals with positive tests actually have HIV.

Exercise 2: Sequential Bayesian Updating

A patient tested positive for HIV. The doctor orders a second test. If we assume the two tests are independent, we can use the posterior from the first test as the new prior.

Round 1 (you already did this)

Prior: P(HIV) = 0.003

Result: Positive

Posterior: P(HIV | +) = _____ (your answer from Exercise 1)

Round 2: Second positive test

New prior: P(HIV) = _____ (posterior from Round 1)

Sensitivity and specificity remain the same: P(B|A) = 0.99, P(B|~A) = 0.05

P(Test+) = P(B|A) × P(A) + P(B|~A) × P(~A)
= 0.99 × _____ + 0.05 × _____
= _____ + _____
= _____

P(HIV | second +) = P(B|A) × P(A) / P(Test+)
= _____ / _____
= _____

Round 3: Third positive test

New prior: P(HIV) = _____ (posterior from Round 2)

P(Test+) = 0.99 × _____ + 0.05 × _____
= _____

P(HIV | third +) = _____ / _____
= _____

Summary

	Prior	Posterior
After 1st positive	0.003	
After 2nd positive		
After 3rd positive		

Q1: How many consecutive positive tests does it take to be more than 95% confident the patient has HIV?

Q2: We assumed each test is independent. Why might this assumption fail if you repeat the exact same test on the same blood sample?

Exercise 3: Mammography Screening

1% of women in a screening population have breast cancer. A mammogram has 90% sensitivity and 92% specificity.

Step 1: List what you know

P(C) = cancer present	_____	*prevalence*
P(~C) = cancer absent	_____	
P(M|C) = mammo+ given cancer	_____	*sensitivity*
P(~M|C) = mammo− given cancer	_____	*false negative rate*
P(M|~C) = mammo+ given no cancer	_____	*false positive rate = 1 − specificity*
P(~M|~C) = mammo− given no cancer	_____	*specificity*

Step 2: P(Mammogram+)?

P(M) = P(M|C) × P(C) + P(M|~C) × P(~C)

= _____ × _____ + _____ × _____

= _____

Mammograms are positive about _____% of the time.

Step 3: P(Cancer | Mammogram+)?

P(C|M) = P(M|C) × P(C) / P(M)

= _____ / _____

= _____

Step 5: Compare all three scenarios

	Prevalence	Sensitivity	Specificity	PPV
Bean exercise	20%	50%	83%	
HIV test	0.3%	99%	95%	
Mammogram	1%	90%	92%	

Q3: The HIV test has better sensitivity AND better specificity than the mammogram. Which has a higher PPV? Why?

Q4: The bean exercise has the worst test (50% sensitivity, 83% specificity) but the highest PPV. What does this tell you about the role of prevalence?

Q5: A woman receives a positive mammogram. In one sentence, what should her doctor tell her about the result?